CROSS-DATASET GENERALIZATION OF ANOMALY DETECTION MODELS FOR PUBLIC SAFETY

Generalization from UCF-Crime to Real-Life Violence Situations

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1. Abstract

Domain shift is a critical obstacle in deploying video anomaly detection systems across real-world settings. This project evaluates how well an LSTM-based anomaly detection model, trained on the UCF-Crime surveillance dataset, transfers to the Real-Life Violence Situations Dataset (RLVD), a collection of handheld camera clips from a different distribution. Without adaptation, the model collapses to chance-level performance on the target domain, achieving an accuracy of 0.50 and a ROC-AUC of 0.49. Applying lightweight linear probing on frozen backbone features recovers substantial performance, raising accuracy to 0.77 and ROC-AUC to 0.83 using fewer than 140 labelled examples. The results demonstrate that deep feature representations generalize across domains even when the decision boundary does not, and that sample-efficient adaptation is sufficient to bridge a significant domain gap in violence detection.

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2. Introduction

Motivation

The proliferation of closed-circuit television (CCTV) infrastructure in public spaces has created both an opportunity and a challenge for automated video analysis. Modern surveillance networks generate footage at a scale that far exceeds the capacity of human operators to monitor in real time. Automated systems capable of detecting anomalous events, such as physical violence, robberies, or accidents, could substantially augment public safety operations by flagging incidents for human review. The need for such systems has driven significant research interest in video anomaly detection, with several large-scale benchmark datasets and deep learning approaches now available.

Despite this progress, a fundamental gap remains between laboratory benchmark performance and practical deployment. Surveillance models are typically trained and evaluated on a single dataset, under the implicit assumption that the deployment environment resembles the training environment. In practice, this assumption rarely holds. Camera hardware, mounting angles, lighting conditions, scene types, and even the definition of what constitutes an anomalous event all vary across deployment sites. A model trained on one corpus may perform well on held-out samples from the same distribution while failing almost entirely when transferred to footage collected under different conditions. This phenomenon is known as domain shift, and it represents one of the most practically significant obstacles to real-world anomaly detection.

Problem Statement

This project investigates domain shift in the context of violence detection for surveillance video. Specifically, it evaluates a pre-trained LSTM-based anomaly detection model, originally developed for the UCF-Crime dataset, when transferred to the Real-Life Violence Situations Dataset (RLVD). UCF-Crime consists of long, untrimmed CCTV recordings captured by stationary institutional cameras, annotated across thirteen anomaly categories. RLVD consists of short, trimmed clips recorded with handheld cameras, depicting either physical violence or non-violent activity in settings such as sports arenas and street scenes. The two datasets differ in camera type, clip duration, annotation granularity, and the semantic scope of what is labelled as violent. Evaluating across this gap provides a demanding but realistic test of cross-dataset generalization.

The core research questions are as follows. First, to what degree does out-of-domain performance degrade relative to in-domain performance? Second, can a lightweight adaptation strategy, specifically linear probing on frozen backbone features, recover meaningful performance using a small number of labelled target-domain examples?

Scope and Contributions

The project makes three primary contributions. First, it establishes an in-domain baseline by evaluating the pre-trained model on the UCF-Crime test split, providing a reference point for the model's intended performance. Second, it quantifies the domain shift by applying the same model, without modification, to the RLVD dataset and documenting the resulting failure mode in detail. Third, it demonstrates that linear probing of the model recovers substantial performance on RLVD, achieving an accuracy of 0.77 and a ROC-AUC of 0.83 with fewer than 140 labelled training examples, compared to an accuracy of 0.50 and a ROC-AUC of 0.49 before adaptation.

An additional methodological contribution concerns the design of the binary scoring function. The model produces a fourteen-class probability distribution, and collapsing this to a binary violence score requires a deliberate design choice. This project documents an initial failed attempt using a selective class-sum approach, diagnoses the cause of the failure (probability mass fragmentation across semantically related classes), and derives a more principled mapping based on the complement of the model's estimated Normal-class probability.

Document Roadmap

The remainder of this report is organized as follows. Chapter 3 reviews related work on video anomaly detection, domain adaptation, and transfer learning with frozen backbones. Chapter 4 describes the two datasets, their properties, and the key differences that give rise to domain shift. Chapter 5 details the model architecture, including the GoogLeNet feature extraction backbone and the LSTM-based classification head. Chapter 6 covers the methodology, encompassing the binary mapping design, the feature extraction pipeline for RLVD, and the linear probing adaptation strategy. Chapter 7 presents the experimental results across all three evaluation phases. Chapter 8 analyses the results and discusses the mechanisms underlying both the domain shift failure and the adaptation recovery. Chapter 9 enumerates the limitations of the study. Chapter 10 concludes with a summary of findings and directions for future work.

3. Related Work

Anomaly Detection in Surveillance Video

Anomaly detection in surveillance video is framed as a weakly supervised learning problem in the most influential work in the field. Sultani, Chen, and Shah (2018) introduced a multiple instance learning (MIL) framework in which training videos are treated as bags of instances, with only video-level labels indicating whether an anomaly is present anywhere in the clip. Under this formulation, a ranking loss encourages the model to assign higher anomaly scores to the most suspicious segments of positive bags (anomalous videos) relative to the highest-scoring segments of negative bags (normal videos). This approach avoids the prohibitive cost of frame-level temporal annotation and enables training on large, weakly labelled corpora. The UCF-Crime dataset, introduced in the same work, became the standard benchmark for this line of research, containing approximately 128 hours of surveillance footage across thirteen anomaly categories.

Subsequent work has extended this foundation in several directions. Zhong et al. (2019) reformulated the problem as a supervised learning task by using a graph convolutional network to generate pseudo frame-level labels from weakly labelled data, improving temporal precision without requiring manual frame annotation. Tian et al. (2021) proposed a two-stream approach combining RGB appearance features with optical flow to capture motion dynamics more explicitly. Despite these advances, the shared limitation across most work in this area is the assumption that training and test data are drawn from the same distribution, an assumption that this project explicitly challenges.

Video Feature Extraction Backbones

The choice of feature extraction backbone has a significant bearing on anomaly detection performance. Early work relied on hand-crafted spatial-temporal descriptors such as improved dense trajectories (Wang and Schmid, 2013). The introduction of deep convolutional approaches, most notably C3D (Tran et al., 2015), allowed the joint learning of spatial and temporal representations from raw video frames. C3D uses three-dimensional convolutions over short temporal windows and produces compact, semantically rich representations that transfer well across video understanding tasks.

Subsequent architectures improved upon C3D in representational capacity. The Inception family of networks, including GoogLeNet (InceptionV1, Szegedy et al., 2015), was designed for image classification but has been widely adopted as a per-frame feature extractor in video pipelines. When applied to individual frames, the 1024-dimensional average-pooling layer output of GoogLeNet captures rich appearance features and, when aggregated over a sequence, provides an effective representation of scene content over time. Two-stream Inflated 3D ConvNet (I3D, Carreira and Zisserman, 2017) further advanced the field by inflating 2D ImageNet-pretrained weights into 3D convolutions, enabling strong spatial-temporal representations with efficient pre-training. The model used in this project employs GoogLeNet per-frame features as its input representation, an approach representative of the generation of pipelines that pre-extract frame-level descriptors and feed them into a sequence model.

Domain Adaptation in Video Understanding

Domain adaptation addresses the problem of transferring a model trained on a labelled source domain to a target domain where labels are scarce or unavailable. In the image domain, methods based on adversarial feature alignment (Ganin and Lempitsky, 2015) and moment matching (Long et al., 2015) have been extensively studied. Applying these techniques to video is complicated by the additional temporal dimension and the higher computational cost of processing sequences.

In the video understanding literature, domain adaptation has been explored primarily in the context of action recognition. Chen et al. (2019) proposed a temporal relational adaptation network that aligns both frame-level and clip-level feature distributions across domains. Munro and Damen (2020) studied adaptation between first-person and third-person video for action recognition, demonstrating that even structural differences in viewpoint introduce distributional shift that standard supervised models cannot bridge. For anomaly detection specifically, cross-dataset adaptation remains relatively underexplored, in part because the availability of labelled anomaly data is limited by design. This project addresses this gap by evaluating cross-dataset transfer between two datasets that differ substantially in camera type, clip structure, and semantic scope.

Adaptation strategies can be broadly categorized as shallow or deep. Deep adaptation methods, such as fine-tuning all or most network layers on target-domain data, typically require large amounts of labelled data in the target domain and are computationally intensive. Shallow adaptation methods, including linear probing and feature-space re-weighting, operate on a fixed frozen backbone and require far fewer labelled examples. These methods are particularly attractive in surveillance applications where annotation is expensive and target-domain data may be limited.

Transfer Learning with Frozen Backbones

The practice of freezing a pre-trained backbone and training only a lightweight classification head on top is well established in image classification and natural language processing. Kumar et al. (2022) showed that fine-tuning a pre-trained model can distort features that were well-suited to the target domain, and that linear probing often provides a more stable starting point when target-domain data is limited. In the video domain, Zhai et al. (2019) demonstrated that features extracted from models pre-trained on large video corpora transfer effectively to downstream tasks even without further fine-tuning of the backbone. The effectiveness of linear probing depends on the degree to which the source and target domains share relevant low-level and mid-level features. When the shared structure is primarily in the feature space rather than the decision boundary, freezing the backbone and learning only a new linear head is sufficient to recover strong performance.

This intuition motivates the adaptation strategy used in this project. The UCF-Crime-trained backbone is expected to encode general motion and appearance features relevant to violence detection, even though its original decision boundary was calibrated to distinguish normal from anomalous surveillance footage rather than to classify violent versus non-violent activity in a different visual domain.

Cross-Dataset Evaluation in Anomaly Detection

Cross-dataset generalization has been studied most extensively in face recognition and person re-identification, where models trained on one collection of identities are evaluated on a disjoint set. In the anomaly detection setting, direct cross-dataset comparisons are less common, partly because datasets differ not only in visual appearance but also in the definition of the target class. Lu et al. (2020) evaluated several anomaly detection models across multiple campus and traffic surveillance datasets and found that performance dropped substantially when models were applied to out-of-distribution footage, even within the same broad domain of fixed-camera surveillance. The present work extends this line of inquiry by crossing a larger domain boundary: from fixed institutional CCTV to handheld consumer video, and from a broad thirteen-category anomaly taxonomy to a binary violence classification task. To the authors' knowledge, no prior work has evaluated the specific UCF-

Crime-to-RLVD transfer investigated here, making the domain shift characterization and lightweight adaptation results a novel empirical contribution.

4. Datasets

UCF-Crime

The UCF-Crime Anomaly Detection Dataset (Sultani et al., 2018) is the primary benchmark for weakly supervised video anomaly detection. It comprises approximately 128 hours of untrimmed surveillance footage collected from CCTV cameras in institutional and public settings. The dataset covers thirteen anomaly categories: Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, and Vandalism. A fourteenth class, Normal, contains footage with no anomalous activity. Videos are long and untrimmed, with anomalous events occupying only a fraction of each clip's duration. This structure reflects realistic surveillance conditions, where incidents are rare relative to the total volume of recorded footage.

Pre-extracted per-frame features are provided with the dataset, computed using GoogLeNet (InceptionV1) with the average-pooling layer as the feature source, yielding a 1024-dimensional vector per frame. These feature files eliminate the need to process raw video during model training and evaluation, which is the format consumed directly by the LSTM-based model used in this project. The dataset also includes a train-test split defined by manifest files. The test split nominally contains 140 anomalous videos and 50 Normal videos, for a total of 190 videos.

For binary classification purposes, the Normal class (index 7) is treated as the NonViolence class, and all thirteen anomaly categories are treated as Violence. This mapping follows directly from the model's training objective, which was to distinguish normal surveillance footage from any anomalous event, regardless of type.

Real-Life Violence Situations Dataset

The Real-Life Violence Situations Dataset (RLVD; Soliman et al., 2019) was constructed to support binary violence detection in consumer video. It contains 1000 Violence clips and 951 NonViolence clips in MP4 format, for a total of 1951 videos. Violence clips depict physical confrontations between individuals, sourced from street fights, brawls, and similar events recorded by bystanders or security personnel with handheld devices. NonViolence clips depict non-violent activity drawn from diverse sources including sports footage, street scenes, and everyday social interactions. All clips are short and trimmed, typically a few seconds to around ten seconds in duration, and carry video-level binary labels with no temporal annotation of event boundaries within a clip.

Unlike UCF-Crime, RLVD does not provide pre-extracted features. For this project, GoogLeNet per-frame features were extracted at runtime using a PyTorch implementation of InceptionV1, matching the feature extraction methodology used for UCF-Crime. A stratified sample of 200 clips (100 Violence, 100 NonViolence) was drawn from the full dataset using a fixed random seed (seed = 42), and 40 frames were sampled uniformly from each clip to produce a feature sequence of shape (40, 1024). Extracted features were cached as NumPy arrays to avoid redundant computation.

The choice of a 200-video sample reflects the hardware constraints of the project (CPU-only environment) and the goals of a proof-of-concept study. The implications of this sampling for metric variance are discussed in Chapter 9.

Comparison of Dataset Characteristics

The two datasets differ along several dimensions that are directly relevant to domain shift. Table 1 summarises the key contrasts.

Table 1. Comparison of UCF-Crime and RLVD dataset characteristics.

Property	UCF-Crime	RLVD
Camera type	Fixed institutional CCTV	Handheld consumer devices
Clip structure	Long untrimmed recordings	Short trimmed clips
Typical clip duration	Minutes to tens of minutes	A few seconds to ~10 seconds
Number of classes	13 anomaly categories + Normal	Violence, NonViolence (binary)
Annotation granularity	Video-level (train); temporal (test)	Video-level only
Pre-extracted features	Yes (GoogLeNet 1024-d)	No (extracted for this project)
Total labelled videos (full corpus)	~1900 (train + test)	1951
Violence semantics	Any anomalous event (including Arson, Shoplifting)	Physical human-to-human confrontation

The most consequential difference for this study is the mismatch in what constitutes Normal footage. The UCF-Crime model learned its concept of normality from long, low-activity CCTV recordings of empty corridors, car parks, and institutional spaces. RLVD NonViolence clips, by contrast, contain dynamic handheld footage of sports events, pedestrian activity, and social gatherings. From the perspective of a model trained on UCF-Crime, this content does not resemble static surveillance normality, which causes the model to assign anomaly scores above the detection threshold to all RLVD clips regardless of their true label. This mechanism is the primary driver of the domain shift documented in this project.

A secondary difference is the semantic scope of the Violence class. Under the binary mapping used in this project, all thirteen UCF-Crime anomaly categories are treated as Violence. This includes categories such as Shoplifting, Arson, and Road Accidents that do not constitute physical violence under RLVD's definition. The model's decision boundary therefore encodes a broader notion of anomaly than the target task requires, introducing a further source of mismatch that cannot be resolved without retraining or relabelling.

5. Model Architecture

Feature Extraction: GoogLeNet

The input representation used by the model consists of per-frame features extracted from video using GoogLeNet (InceptionV1; Szegedy et al., 2015). GoogLeNet is a deep convolutional neural network originally developed for the ImageNet Large Scale Visual Recognition Challenge, notable for its use of parallel convolutional branches (Inception modules) that capture spatial structure at multiple scales within a single layer. For the purposes of this project, GoogLeNet is used as a fixed feature extractor rather than a trainable component. The output of the global average-pooling layer, which produces a 1024-dimensional vector summarising the spatial content of each frame, is used as the frame-level descriptor.

To represent a video as a fixed-size sequence, 40 frames are sampled uniformly across the clip's duration. For long UCF-Crime recordings, this uniform sampling covers the full extent of the video, including periods before and after any anomalous event. For short RLVD clips of a few seconds, the same 40-frame sampling covers the entire clip more densely. Each video is thus represented as a matrix of shape (40, 1024), which serves as the input to the sequence model.

Sequence Model: LSTM-based Classifier

The core classification network takes the (40, 1024) feature sequence as input and processes it through the following layer stack:

```
LSTM(1024) → Dense(512) → Dropout → Dense(32) → Dropout → Dense(14)
```

The LSTM layer (Long Short-Term Memory; Hochreiter and Schmidhuber, 1997) contains 1024 hidden units and processes the sequence of frame-level features, capturing temporal dependencies across the 40 frames. LSTM units are well suited to this task because they can selectively retain or discard information over time through their gating mechanisms, allowing the model to attend to relevant frames within a long sequence even when the anomalous event is brief.

The LSTM output is passed to a fully connected Dense layer with 512 units, followed by a Dropout layer for regularisation. A second Dense layer with 32 units and a further Dropout layer follow. The final Dense layer produces 14-dimensional logits, one per UCF-Crime class. A softmax activation converts these logits to a probability distribution over the fourteen classes.

The model was trained on the UCF-Crime training split under a weakly supervised multiple instance learning regime, consistent with the framework of Sultani et al. (2018). Under this training procedure, the model learns to distinguish normal surveillance footage from anomalous events without requiring frame-level temporal annotations.

Binary Violence Score

At inference time, the 14-class probability distribution is collapsed to a scalar binary score. As discussed in Chapter 6, the score is defined as one minus the probability assigned to the Normal class (index 7):

```
score = 1 - P(Normal)
```

A score above 0.5 is classified as Violence; at or below 0.5 is classified as NonViolence. This mapping treats the model's own estimate of normality as the sole criterion for the binary decision, which is more faithful to the model's training objective than summing probabilities from a selected subset of violent-category classes.

Feature Layer Selection for Adaptation

When adapting the model to the RLVD domain via linear probing, the choice of which layer's activations to use as the feature representation is consequential. The model offers three candidate internal representations: the LSTM output (1024-dimensional), the first Dense layer output (512-dimensional), and the second Dense layer output (32-dimensional).

The Dense(512) layer was selected for the following reasons. The LSTM output, while rich, is the highest-dimensional option and precedes any task-specific compression; it contains substantial redundancy and is closest to raw temporal aggregation. The Dense(32) layer, by contrast, has been compressed so aggressively that it sits immediately before the 14-class classification head, meaning its representation is already shaped by the UCF-Crime class structure rather than general visual features. The Dense(512) layer occupies a middle ground: it retains sufficient representational capacity (512 dimensions) to support a new linear classifier, while having undergone enough task-relevant compression to encode discriminative features about scene content and activity type. Empirically, this layer is expected to encode information about motion patterns and visual dynamics that are transferable to the RLVD task, even if the downstream class boundary needs to be relearned.

Note on Deprecated Model Artefacts

The project directory contains two additional model-related files that are not used in this study. The file `model.json` in the project root describes a Keras 1.1.0 binary MIL model that takes 4096-dimensional C3D features as input. This is a different architecture, trained under a different framework, and is incompatible with the GoogLeNet feature pipeline. A `saved_model.pb` file at the project root level is similarly deprecated. The model used throughout this project is the TensorFlow SavedModel located at `dataset/UCF-Crime/Model_Trained/Model_Trained/saved_model/my_model`, which implements the LSTM stack described above and accepts 1024-dimensional GoogLeNet features.

6. Methodology

Binary Mapping Design

The UCF-Crime model produces a 14-class softmax probability distribution at inference time. Because the downstream evaluation task is binary (Violence vs. NonViolence), a mapping from this distribution to a scalar score is required. The design of this mapping has a substantial effect on classification performance and is not trivially determined.

Initial Approach and Failure

The intuitive first approach was to select the subset of UCF-Crime classes that correspond semantically to physical violence and sum their probabilities:

```
score = P(Abuse) + P(Assault) + P(Fighting) + P(Robbery) +
P(Shooting)
```

This formulation treats the five most violence-proximate categories (indices 0, 3, 6, 9, 10) as the positive class and everything else as negative. Applied to the UCF-Crime test set, this mapping produced the following results: accuracy 0.80, F1 (Violence) 0.00, ROC-AUC 0.82.

The F1 score of zero indicates that the model predicted zero videos as Violence under this scheme, classifying every video as NonViolence. The ROC-AUC of 0.82 simultaneously confirms that the model's scores do rank violent videos above non-violent ones; the failure is not one of discriminability but of calibration relative to the 0.5 threshold.

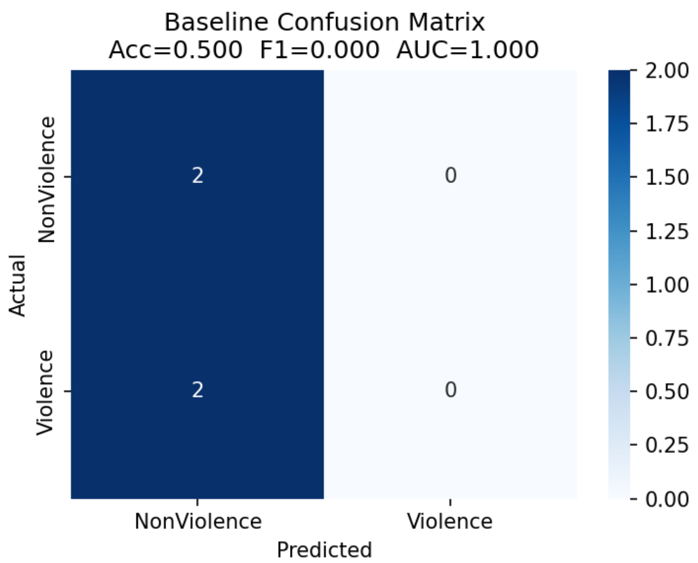

Figure 1. Confusion matrix for the initial 5-class sum scoring approach, showing complete failure to predict the Violence class (F1 = 0.00).

Diagnosis: Probability Mass Fragmentation

The cause of this failure is a structural property of the 14-class softmax distribution. Because all fourteen class probabilities must sum to 1.0, the five selected violence classes compete for probability mass against nine other classes. Even when the model confidently identifies a video as anomalous, that confidence is distributed across multiple anomaly categories. A typical violent video might receive probabilities of roughly 0.18, 0.09, 0.12, 0.06, and 0.05 for the five selected classes, summing to approximately 0.50 and barely meeting the detection threshold. In practice, this fragmentation causes the summed score to remain below 0.5 for most videos, resulting in a systematic bias toward the NonViolence prediction regardless of actual content.

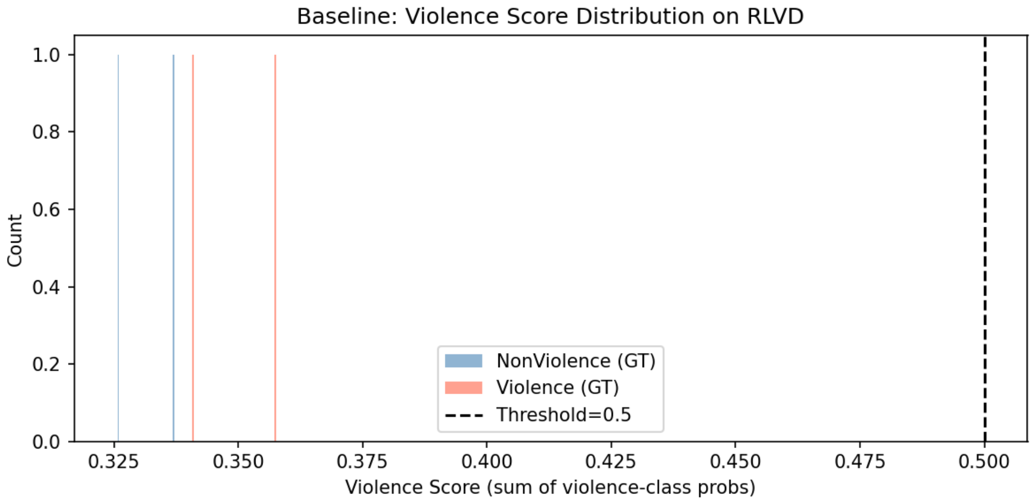

Figure 2. Violence score distribution under the 5-class sum mapping. All scores cluster below the 0.5 threshold, explaining the F1 = 0.00 result.

Revised Mapping

The correct perspective is that the model was not trained to detect physical violence specifically; it was trained to distinguish normal surveillance footage from any anomalous event. The model's learned decision boundary is between the Normal class and all thirteen anomaly categories collectively. The most principled binary conversion therefore mirrors this boundary directly:

```
score = 1 - P(Normal)
```

Under this mapping, NonViolence corresponds exclusively to the Normal class (index 7), and Violence corresponds to the union of all thirteen anomaly categories. The score is the model's own estimate of the probability that the footage contains something anomalous, which is precisely what the model was trained to produce. This approach does not require selecting a subset of classes or making assumptions about which anomaly categories are most relevant; it delegates the decision entirely to the model's learned representation of normality.

This mapping was adopted for all experiments in this project. Its implications for metric interpretation, particularly the inverted AUC on the UCF-Crime test set, are discussed in Chapter 8.

Feature Extraction Pipeline for RLVD

The UCF-Crime model expects per-frame GoogLeNet features as input, but RLVD does not provide pre-extracted features. A feature extraction pipeline was therefore implemented using a PyTorch implementation of GoogLeNet (InceptionV1), configured to produce 1024-dimensional average-pooling layer outputs consistent with the UCF-Crime feature format.

A stratified sample of 200 clips was drawn from the 1951-video RLVD corpus, comprising 100 Violence and 100 NonViolence clips, selected using a fixed random seed (seed = 42) to ensure reproducibility. From each clip, 40 frames were sampled at uniform intervals across the clip's duration using integer-spaced indices. Each frame was resized and normalised according to the ImageNet preprocessing convention expected by GoogLeNet (centre crop to 224 x 224, mean subtraction, standard deviation normalisation). The resulting per-frame feature vectors were stacked into a matrix of shape (40, 1024) for each video.

Extracted features were saved as NumPy `.npy` files, one per video, to avoid redundant computation across evaluation runs. The extraction process ran on CPU-only hardware and required approximately 30 minutes for the full 200-video sample.

Adaptation Strategy: Linear Probing

To adapt the model to the RLVD domain, a linear probing strategy was employed. The full backbone (GoogLeNet feature extractor and the LSTM-based classification network) was frozen; no weights were updated during adaptation. Instead, the 512-dimensional activations of the first Dense layer (Dense(512)) were extracted for each video in the 200-video RLVD sample by running a forward pass through the network up to and including that layer.

The 200 samples were partitioned into training, validation, and test splits in a 70/15/15 ratio (140 training, 30 validation, 30 test), using stratified sampling to preserve class balance across all three splits. Feature vectors were standardised using a StandardScaler fitted on the training split only, and the same scaler was applied to the validation and test splits. This prevents any information from the evaluation data from influencing the feature normalisation.

A logistic regression classifier (scikit-learn LogisticRegression, solver = lbfgs, C = 1.0) was fitted on the 140 standardised training feature vectors. The regularisation strength C = 1.0 was the default value and was not tuned on the validation set, reflecting the limited size of the available training data and the goal of demonstrating adaptation rather than optimising performance. The trained classifier was then evaluated on the 30-video test split. This evaluation constitutes Experiment 3 and is reported in Chapter 7.

The validation split was retained to support potential hyperparameter search in future work, but was not used for any decisions in the experiments reported here. All adaptation results therefore reflect performance on a held-out test set that the classifier never observed during fitting.

7. Experiments and Results

This chapter reports the outcomes of the three experimental phases conducted in this project: an in-domain baseline evaluation on UCF-Crime, an out-of-domain evaluation on RLVD, and an evaluation of the adapted model on a held-out RLVD test split. All experiments use the binary scoring function `score = 1 - P(Normal)` with a classification threshold of 0.5, as derived in Chapter 6.

Experiment 1: UCF-Crime In-Domain Baseline

The first experiment establishes a reference point by evaluating the pre-trained model on its native domain. The evaluation set consists of 190 videos drawn from the UCF-Crime test split: 140 anomaly videos (across the thirteen anomaly categories) and 50 Normal videos.

Under the `1 - P(Normal)` mapping, the model achieved an accuracy of 0.7368, an F1 score of 0.8485 for the Violence class, a precision of 0.7368, a recall of 1.0000, and a ROC-AUC of 0.2197. The recall of 1.0 indicates that all 140 anomaly videos were correctly flagged as Violence. However, the precision of 0.7368 reveals that the 50 Normal videos were also predicted as Violence in their entirety, since precision under perfect recall and a positive class size of 140 implies that all videos, anomaly and Normal alike, received a positive prediction. The very low ROC-AUC of 0.2197, substantially below the chance level of 0.5, indicates that the model's continuous violence scores rank Normal videos higher than anomaly videos on average, an inversion that is explained in detail in Chapter 8.

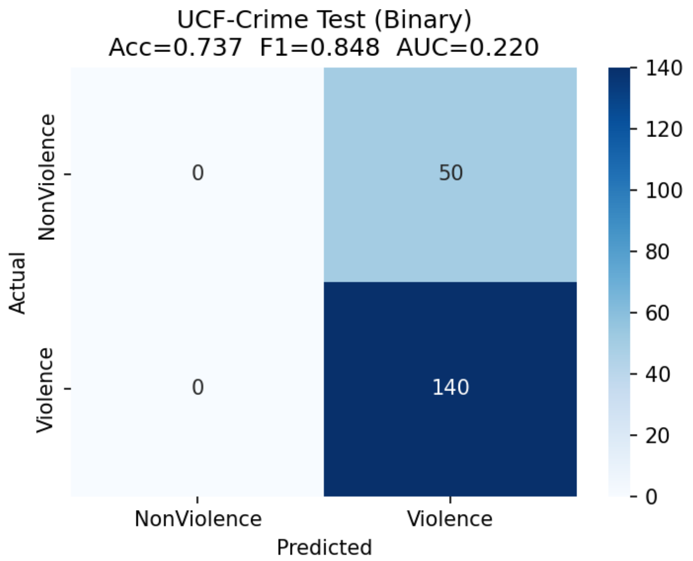

Figure 3. Confusion matrix for the UCF-Crime in-domain baseline evaluation (190 videos).

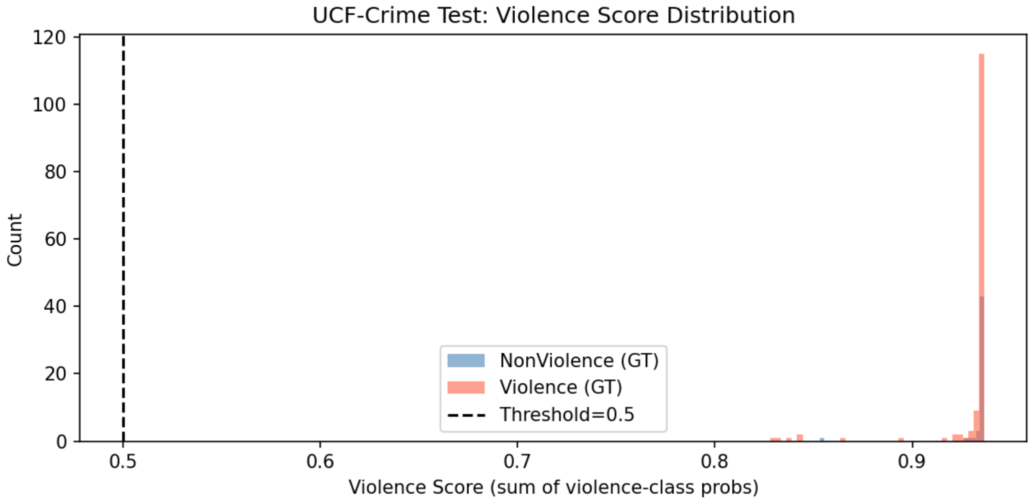

Figure 4. Violence score distribution on the UCF-Crime test set. Note: the x-axis label reflects an earlier script version; scores shown are computed using the 1 − P(Normal) mapping.

Experiment 2: RLVD Out-of-Domain Evaluation

The second experiment applies the same pre-trained model, without any modification or retraining, to the stratified 200-video RLVD sample described in Chapter 6 (100 Violence and 100 NonViolence clips, seed = 42). This experiment constitutes the principal evidence of domain shift in this study.

The out-of-domain results were: accuracy 0.5000, F1 (Violence) 0.6667, precision 0.5000, recall 1.0000, and ROC-AUC 0.4878. As in Experiment 1, recall reached 1.0, meaning every Violence clip was correctly identified. However, precision fell to exactly 0.5000, the level expected from random guessing on a balanced binary task. Taken together with the recall of 1.0 and the balanced sample composition (100 Violence, 100 NonViolence), these figures indicate that the model classified all 200 RLVD clips as Violence, regardless of their true label. NonViolence clips were never predicted correctly: precision, recall, and F1 for the NonViolence class are all 0.00. The ROC-AUC of 0.4878, indistinguishable from the chance level of 0.5, confirms that the model's underlying continuous scores carry no meaningful ranking signal between the two classes on RLVD; the score distributions for Violence and NonViolence clips overlap almost completely.

This pattern represents a complete prediction collapse: the model's output, in the target domain, degenerates into a constant positive prediction that happens to achieve 100 percent recall purely as a side effect of never predicting the negative class.

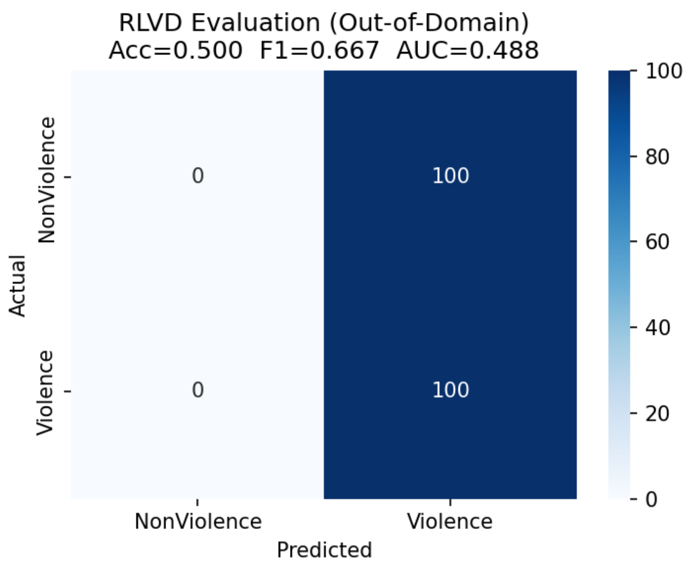

Figure 5. Confusion matrix for the RLVD out-of-domain evaluation (200 videos), showing the complete prediction collapse.

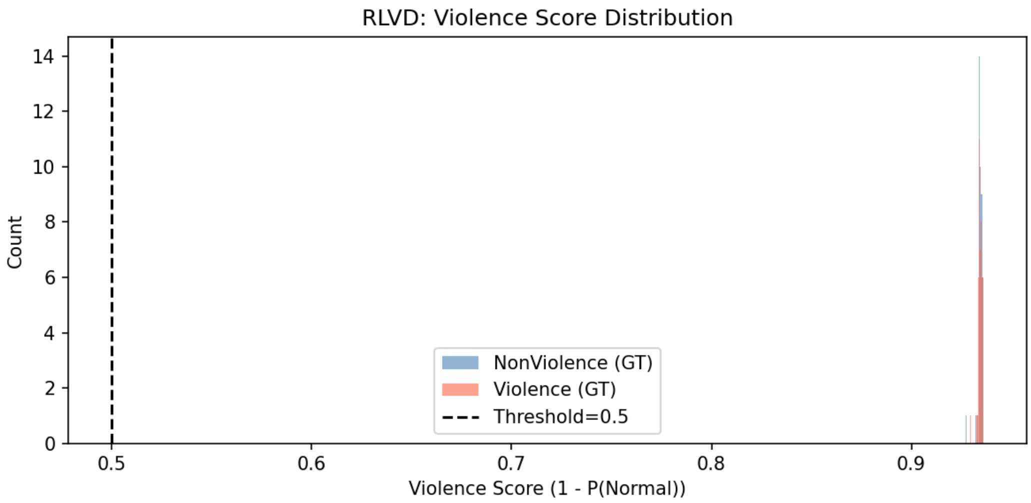

Figure 6. Violence score distribution on RLVD (out-of-domain). Both classes cluster near the upper end of the score range, showing complete overlap.

Experiment 3: Adapted Model on RLVD Test Split

The third experiment evaluates the linear probing adaptation described in Chapter 6. A logistic regression classifier was trained on the 512-dimensional Dense-layer activations of 140 RLVD training videos (the 70 percent training partition of the 200-video sample) and evaluated on a held-out test split of 30 videos (the 15 percent test partition), which the classifier never observed during fitting.

The adapted model achieved an accuracy of 0.7667, an F1 score of 0.7742 for the Violence class, a precision of 0.7500, a recall of 0.8000, and a ROC-AUC of 0.8311. Unlike the out-of-domain baseline, the adapted model achieves a balanced precision-recall profile rather than a degenerate constant prediction, and its ROC-AUC indicates that its continuous scores now meaningfully separate the two classes.

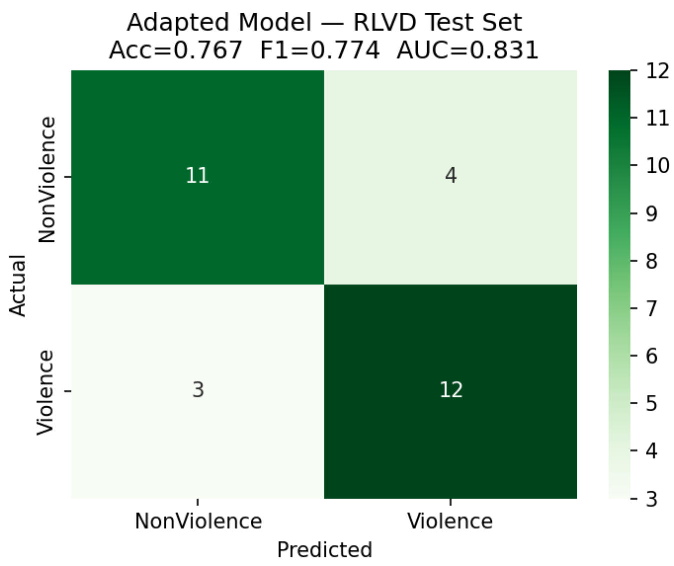

Figure 7. Confusion matrix for the adapted model on the RLVD test split (30 videos).

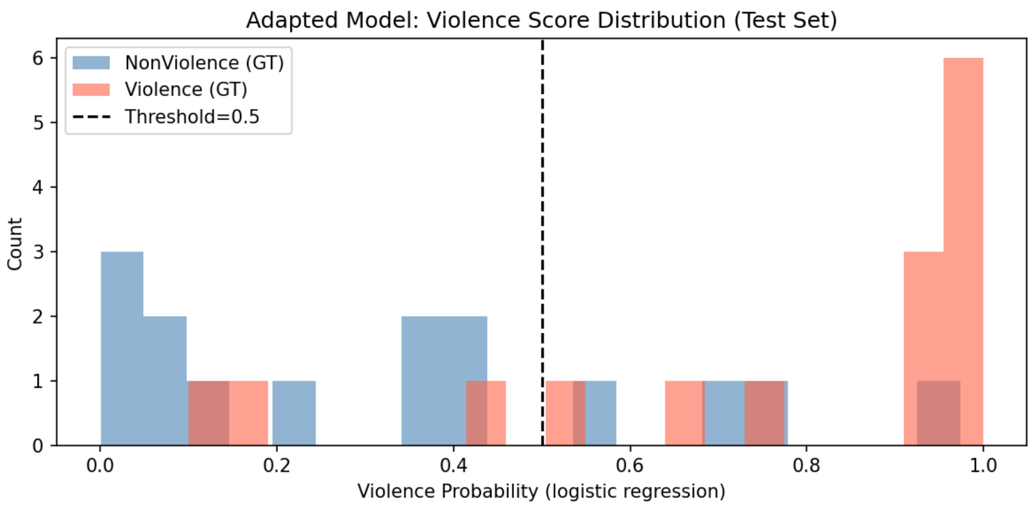

Figure 8. Violence score distribution for the adapted model on the RLVD test split, showing class separation after linear probing.

Master Comparison

Table 2 consolidates the results of all three experiments across the five evaluation metrics.

Table 2. Comparison of model performance across the three experimental phases.

Phase	Dataset	Accuracy	F1 (Violence)	Precision	Recall	ROC-AUC
Baseline (in-domain)	UCF-Crime	0.7368	0.8485	0.7368	1.0000	0.2197
Baseline (out-of-domain)	RLVD	0.5000	0.6667	0.5000	1.0000	0.4878
Adapted (linear probing)	RLVD test	0.7667	0.7742	0.7500	0.8000	0.8311

The transition from the first row to the second row quantifies the cost of domain shift: accuracy falls by 23.7 percentage points, F1 falls by 18.2 percentage points, and ROC-AUC, despite remaining low in both cases, shifts from a strongly inverted ranking on UCF-Crime to a near-random ranking on RLVD. The transition from the second row to the third row quantifies the benefit of adaptation: relative to the unadapted out-of-domain baseline, accuracy rises by 26.7 percentage points, F1 rises by 10.7 percentage points, precision rises by 25.0 percentage points, and ROC-AUC rises by 34.3 percentage points. Recall decreases from 1.0 to 0.8, which is an expected consequence of the model moving away from a degenerate all-positive prediction strategy toward a genuine decision boundary that can also identify the NonViolence class.

These results jointly demonstrate that the pre-trained model suffers a severe and well-characterised collapse when transferred to RLVD without adaptation, and that a lightweight, frozen-backbone linear probing strategy using fewer than 140 labelled examples is sufficient to recover the majority of the lost discriminative capacity. The mechanisms underlying both the collapse and the recovery are examined in Chapter 8.

8. Analysis and Discussion

The results presented in Chapter 7 raise several questions that require closer examination. Why does the model collapse into a constant Violence prediction on RLVD? Why does the ROC-AUC behave so differently on the two datasets, registering a strongly inverted value of 0.22 on UCF-Crime and a near-random value of 0.49 on RLVD, despite both indicating poor discrimination by conventional standards? And why does a simple linear classifier, trained on fewer than 140 examples, recover so much of the lost performance? This chapter addresses each of these questions in turn.

The Mechanism Behind the NonViolence Collapse

The defining characteristic of the out-of-domain results is that the model predicts every RLVD clip, violent or not, as Violence. This is not a marginal miscalibration but a complete failure to ever produce a NonViolence output. The explanation lies in what the model actually learned to recognise as "Normal."

During training on UCF-Crime, the Normal class consisted of long, low-activity surveillance recordings: static cameras observing corridors, car parks, and other institutional spaces with minimal movement. The model's internal representation of normality is therefore tied to visual stillness, fixed viewpoints, and low scene dynamics. The probability P(Normal), and consequently the score 1 - P(Normal), reflects how closely a video resembles this specific notion of quiet, static surveillance footage, rather than a general notion of "nothing bad is happening."

RLVD's NonViolence clips do not resemble this notion at all. They are short, handheld recordings of sports matches, pedestrian street scenes, and social gatherings: visually dynamic content captured from moving viewpoints. To the model, this footage looks nothing like the Normal class it was trained on. Consequently, P(Normal) remains low for these clips, the violence score 1 - P(Normal) remains above the 0.5 threshold, and the clips are classified as Violence. The model is not failing to recognise violence; it is failing to recognise non-violence, because its definition of "non-anomalous" is inseparably bound to a visual style that RLVD does not contain. Put differently, the model cannot distinguish "dynamic but non-violent" footage from "genuinely violent" footage, because neither matches the static surveillance distribution it was trained on.

The confusion matrix (Figure 5) and score distribution plot (Figure 6) from Experiment 2 illustrate this collapse: every clip scores above the 0.5 threshold, with Violence and NonViolence distributions entirely overlapping.

The Frame Dilution Effect in UCF-Crime

A related but distinct mechanism explains the model's behaviour on its own native test set, where the ROC-AUC of 0.2197 indicates that the ranking of scores is, if anything, inverted relative to the true labels.

UCF-Crime videos are long and untrimmed, and the anomalous event within an anomaly video typically occupies only a small fraction of the total duration. When 40 frames are sampled uniformly across such a video, the majority of sampled frames depict ordinary activity rather than the anomaly itself. As a result, the aggregated representation seen by the LSTM is dominated by normal-looking content, and P(Normal) for an anomaly video can remain comparatively high. The anomaly signal is, in effect, diluted across the sequence.

At the same time, the videos used as the Normal test set in this evaluation are drawn from a distinct subset (event-recognition clips) that tend to be shorter and more visually dynamic than the quiet institutional footage that dominates the Normal training distribution. These clips, despite being labelled Normal, do not resemble the model's learned concept of normality any more than the anomaly clips do, and in some cases less so.

The combined effect is a ranking inversion: long anomaly videos, with their anomalies diluted across many normal-looking frames, can receive higher P(Normal) scores than the comparatively dynamic Normal test clips. This produces an AUC below the chance level of 0.5. Critically, this inversion is a property of the test-set composition and the sampling procedure interacting with the model's learned representation; it does not indicate that the model cannot identify anomalies in general; recall on the anomaly class remains 1.0, confirming that the model reliably flags anomaly videos as such even if its ranking of Normal videos is unreliable.

Reconciling the Two AUC Values

The AUC values of 0.22 on UCF-Crime and 0.49 on RLVD might appear to describe similar degrees of failure, but they reflect different underlying phenomena, and reconciling them clarifies the nature of the domain shift.

On UCF-Crime, the AUC is substantially below 0.5, indicating an active inversion: the model's scores rank many Normal videos above many anomaly videos. This inversion arises specifically from the frame dilution effect described above, a consequence of long, untrimmed video structure interacting with uniform frame sampling. On RLVD, the AUC is close to 0.5, indicating an absence of ranking signal rather than an inversion: Violence and NonViolence clips receive scores that are statistically indistinguishable from one another. There is no dilution problem on RLVD, since clips are short and trimmed, so the inversion mechanism does not apply. Instead, the flat AUC reflects the fact that the model's notion of "anomalous" and "normal" simply does not map onto the Violence and NonViolence categories in this new visual domain.

In short, the UCF-Crime AUC reflects a structural artefact of long-video evaluation, while the RLVD AUC reflects a genuine absence of transferable discriminative signal under the original decision boundary. The two numbers are not contradictory; they are symptoms of two different failure modes that happen to produce numerically similar deviations from the ideal AUC of 1.0.

Score Distributions Before and After Adaptation

Examination of the distribution of violence scores, visualised in the figures below (see Figure 6 and Figure 8), provides further insight into the nature of the domain shift and the effect of adaptation.

Prior to adaptation, the score distributions for RLVD Violence and NonViolence clips overlap almost completely, both clustering near or above the 0.5 threshold. This overlap is the direct visual signature of the AUC value close to 0.5: there is no score range in which one class predominates over the other, and any threshold choice would produce similarly poor separation. This is consistent with the conclusion that the pre-trained model's scores carry no class-relevant information in the target domain.

This is visible in Figure 6, where both class distributions occupy the same region near the top of the score range.

Following linear probing adaptation, the score distributions for the two classes separate substantially, with NonViolence clips concentrating at lower scores and Violence clips concentrating at higher scores, though some overlap remains given the small sample size. This separation is the direct visual counterpart of the ROC-AUC improvement from 0.4878 to 0.8311, and it demonstrates that a meaningful decision boundary between the two classes exists within the RLVD feature space, even though the original model could not locate it.

Figure 8 shows this separation: NonViolence clips concentrate at lower scores and Violence clips at higher scores after the linear probing step.

Why Linear Probing on Dense(512) Succeeds

The success of the adaptation strategy, recovering an accuracy gain of 26.7 percentage points and an AUC gain of 34.3 percentage points using fewer than 140 labelled examples and no backbone retraining, indicates that the underlying representation learned on UCF-Crime retains substantial value for the RLVD task. If the GoogLeNet and LSTM backbone produced features that were entirely uninformative about violence in the new domain, no linear classifier operating on those features could achieve such a separation, regardless of how it was trained.

The most plausible explanation is that the Dense(512) activations encode general motion and appearance characteristics, such as the presence of rapid movement, physical contact, crowd density, and scene composition, that correlate with violent activity across visual domains, even though the model's final classification head was calibrated to a UCF-Crime-specific notion of normality. The backbone, in other words, has already done much of the representational work required to distinguish violent from non-violent content; what was missing was a decision boundary appropriate to the new domain's class distribution.

This interpretation is consistent with the broader transfer learning literature discussed in Chapter 3, which holds that frozen-backbone linear probing is most effective when the source and target domains share structure at the feature level, even if their label distributions and decision boundaries differ. The results of this project suggest that UCF-Crime and RLVD share exactly this kind of feature-level structure, and that the principal barrier to direct transfer was not a deficiency in the learned representation but a mismatch between the original decision rule and the requirements of the new task.

9. Limitations

This chapter enumerates the principal limitations of the study. Each limitation is described with reference to its scope and its effect on the conclusions that can be drawn.

Small RLVD Evaluation Sample

The full RLVD corpus contains 1951 videos, but only 200 were used in this study. This sampling was motivated by the CPU-only hardware environment, which made feature extraction and evaluation of the full corpus impractical within the project timeline. While the 200-video sample was stratified to ensure equal class representation, a sample of this size produces metrics with non-trivial variance. Performance estimates based on 100 videos per class are susceptible to fluctuation from individual misclassifications; a change of three correct predictions on a 30-video test split, for example, represents a 10 percentage point change in accuracy. The reported metrics should therefore be understood as point estimates with uncertainty, rather than as precise characterisations of model performance on the full RLVD distribution.

Small Adaptation Training Set

The linear probing classifier was trained on 140 RLVD videos (the 70 percent training partition of the 200-video sample). While this proved sufficient to recover substantial performance in the test evaluation, the small training set limits the generalisability of the adapted classifier. The learned decision boundary may be sensitive to the specific videos that happened to be included in the training split, and performance on a larger or differently sampled RLVD evaluation could differ meaningfully. No hyperparameter tuning was performed on the logistic regression regularisation strength (C = 1.0) due to the limited sample size, which means the classifier may not be optimally configured for this data regime.

Semantic Mismatch Between Violence Labels

The binary mapping used in this project treats all thirteen UCF-Crime anomaly categories as Violence, including Arson, Shoplifting, Stealing, Road Accidents, Vandalism, and Arrest. These categories do not correspond to physical human-to-human confrontation, which is precisely what RLVD's Violence class is defined to capture. The model's internal representations and decision boundaries were therefore shaped by a broader and more heterogeneous notion of "anomaly" than the target task requires. This mismatch is an inherent consequence of adapting a multi-class anomaly detection model to a binary violence classification task without retraining the backbone. The degree to which this semantic gap degrades performance, relative to a model trained specifically on physical violence, cannot be quantified from the current experiments alone.

CPU-Only Hardware

All feature extraction, model inference, and adaptation in this project ran on CPU hardware without GPU acceleration. This constraint limited the practical scale of experiments in two ways. First, it made full-corpus RLVD evaluation impractical, necessitating the 200-video sample discussed above. Second, it precluded exploration of deeper adaptation strategies, such as fine-tuning the LSTM layers or the Dense(512) layer on RLVD data, which would have been substantially slower without GPU support. The study is therefore limited to the least computationally intensive form of adaptation (linear probing on frozen features), and comparisons against more resource-intensive strategies, such as partial or full fine-tuning, are not available.

Video-Level Labels Only for RLVD

RLVD provides only video-level binary labels, with no temporal annotation indicating when within a clip the violent activity occurs. This means that the evaluation reported here measures clip-level classification accuracy rather than temporal localisation performance. The model produces a single binary prediction per clip, and there is no way to assess whether it correctly identifies the temporal extent of violent activity within a clip or merely responds to some global visual feature correlated with violence. For deployment in real surveillance systems, where temporal localisation is often as important as detection, this limitation is significant. Temporal annotation of at least a subset of RLVD clips would be required to evaluate localisation performance.

10. Conclusion

Summary of Findings

This project set out to characterise the degree to which an LSTM-based anomaly detection model trained on UCF-Crime generalises to the Real-Life Violence Situations Dataset, and to evaluate whether lightweight adaptation could recover discriminative performance without retraining the backbone. The findings can be summarised in three main points.

First, the domain shift between UCF-Crime and RLVD is severe. When applied without modification to RLVD, the model collapses into a constant Violence prediction, classifying all 200 evaluation clips as violent regardless of their true label. Accuracy falls to 0.50 and ROC-AUC falls to 0.49, both consistent with chance-level performance. The failure is well-characterised: it stems from a mismatch between the model's learned notion of "Normal" (static institutional CCTV footage) and the visual content of RLVD NonViolence clips (dynamic handheld video), rather than from any deficiency in the backbone's representational capacity.

Second, the choice of binary mapping is a non-trivial methodological decision that can determine whether a model produces meaningful output at all. The initial class-sum approach, which assigned Violence scores by summing probabilities over five semantically relevant UCF-Crime categories, resulted in F1 = 0.00 despite a ROC-AUC of 0.82, because probability mass fragmentation across the 14-class softmax prevented the summed score from crossing the detection threshold. Replacing this with the complement of the Normal-class probability, a mapping that reflects the model's actual training objective, restored the Violence class F1 to 0.85 on UCF-Crime. This finding illustrates that the design of the output scoring function deserves the same scrutiny as the model architecture itself.

Third, frozen-backbone linear probing is an effective and sample-efficient adaptation strategy for this domain gap. Training a logistic regression classifier on the Dense(512) activations of 140 RLVD videos raised accuracy from 0.50 to 0.77 and ROC-AUC from 0.49 to 0.83, representing gains of 26.7 and 34.3 percentage points respectively. The adapted model achieves a balanced precision-recall profile and meaningful class separation, in contrast to the degenerate all-positive baseline. These results confirm that the backbone already encodes features relevant to violence detection in the new domain; only the decision boundary needed to be relearned.

Practical Insights

Two practical implications follow from these findings. For practitioners deploying surveillance anomaly detection models across environments, the results highlight that domain shift can manifest as a silent but total failure: the model continues to produce outputs and achieve non-trivial scores on some metrics (such as recall) while providing no useful discrimination. Validation on a representative sample of the target environment before deployment is therefore important, not only to measure aggregate accuracy but to verify that both classes are being predicted.

The success of linear probing with a small adaptation set also suggests a viable deployment workflow for resource-constrained settings. Collecting and labelling a few hundred clips from the target environment, extracting features from a frozen backbone, and fitting a simple linear classifier is computationally inexpensive and can be accomplished without GPU hardware. This approach does not require access to the original training data or any modification of the pre-trained weights, making it compatible with black-box deployment scenarios where only inference access to the model is available.

Requirements for Production Deployment

The adapted classifier demonstrated in this project should be considered a proof of concept rather than a production-ready system. Deployment in a real surveillance environment would require several additional steps. The adaptation training set of 140 videos is small, and a more robust classifier would benefit from several hundred to several thousand labelled examples drawn from the specific deployment context. Additionally, the current adaptation uses only RLVD videos as the NonViolence class, which may not adequately represent the visual distribution of the actual target site. Collecting target-site-specific Normal footage, including the static surveillance content that the original model expects, would allow the linear classifier to learn a more precise boundary between the relevant classes. Temporal localisation capability, absent in the current formulation, would also be required for real-time alerting systems that need to identify the onset and duration of incidents within longer recordings.

Future Work

Several directions present themselves for extending this work. Gradual unfreezing, in which successive layers of the backbone are fine-tuned on increasing amounts of target-domain data, could bridge the remaining performance gap beyond what linear probing achieves, at the cost of greater computational expense and a higher risk of catastrophic forgetting. Data augmentation on the RLVD training split, through spatial transformations, temporal subsampling variations, or synthetic minority-class examples, could partially compensate for the small adaptation set and reduce sensitivity to individual training samples.

Multi-dataset training, in which the backbone is trained jointly on UCF-Crime and RLVD from the outset, would remove the domain shift problem at its source by exposing the model to the target distribution during representation learning. This approach would require matched annotation conventions across datasets or a multi-task loss formulation. Finally, evaluating the adapted model on a third, unseen dataset, drawn from neither UCF-Crime nor RLVD, would provide a more stringent test of whether the adaptation procedure produces genuinely general representations or merely overfits to the specific RLVD visual distribution. Such a three-dataset evaluation would constitute a stronger empirical foundation for claims about cross-domain robustness in surveillance violence detection.

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